READING NOTES

PURNIMA TIWARI

Contents

1. Introduction

The motivation behind this running record of reading notes on mathematics is to gather the ongoing observations, ideas and questions which arise during my study. The entries are primarily for my own understanding, but I hope they may also be of interest to others who enjoy exploring mathematics. As my understanding evolves, some notes may be revised in the future.

2. Notes

28th Aug, 2026. Let $S^1 = \{z \in \mathbb{C} : |z| = 1\}$, with basepoint 1. We prove that

$$\pi_1(S^1, 1) \cong \mathbb{Z}. \tag{1}$$

For each $n \in \mathbb{Z}$, define $\gamma_n : [0,1] \to S^1$, $\gamma_n(t) = e^{2\pi i n t}$.

Since

$$\gamma_n(0) = 1 = \gamma_n(1), \tag{2}$$

γ_n is a loop based at 1.

Define

$$\Phi : \mathbb{Z} \to \pi_1(S^1, 1), \qquad \Phi(n) = [\gamma_n]. \tag{3}$$

Let

$$p : \mathbb{R} \to S^1, p(x) = e^{2\pi i x}. \tag{4}$$

Every loop $\alpha : [0,1] \to S^1$ based at 1 has a unique lift

$$\widetilde{\alpha} : [0,1] \to \mathbb{R} \tag{5}$$

such that $\widetilde{\alpha}(0) = 0$.

Since $\alpha(1) = 1$, $\quad e^{2\pi i \widetilde{\alpha}(1)} = 1$, and hence $\widetilde{\alpha}(1) \in \mathbb{Z}$.

Define

$$W : \pi_1(S^1, 1) \to \mathbb{Z}, \qquad W([\alpha]) = \widetilde{\alpha}(1). \tag{6}$$

This is well-defined since based-homotopic loops have lifts with the same endpoint.

For γ_n, the lift satisfying $\widetilde{\gamma_n}(0) = 0$ is $\widetilde{\gamma_n}(t) = nt$.

Therefore $W(\Phi(n)) = W([\gamma_n]) = \widetilde{\gamma_n}(1) = n$.

Hence $W \circ \Phi = \mathrm{id}_{\mathbb{Z}}$.

Conversely, let $[\alpha] \in \pi_1(S^1, 1)$, and suppose $W([\alpha]) = n$.

Then $\widetilde{\alpha}$ is a path in $\mathbb{R}$ from 0 to n.

Define

$$H : [0,1] \times [0,1] \to \mathbb{R} \text{ by } H(s,t) = (1-s)\widetilde{\alpha}(t) + snt. \tag{7}$$

Then $H(0,t) = \widetilde{\alpha}(t)$, $H(1,t) = nt$, and, for every $s \in [0,1]$, $H(s,0) = 0$, $H(s,1) = n$.

Thus H is a homotopy relative to the endpoints between $\widetilde{\alpha}$ and the path $t \mapsto nt$.

Applying p, we obtain a based homotopy between α and γ_n. Hence $[\alpha] = [\gamma_n] = \Phi(n)$.

Therefore $\Phi \circ W = \mathrm{id}_{\pi_1(S^1,1)}$.

Consequently, Φ and W are inverse isomorphisms, and hence

$$\pi_1(S^1, 1) \cong \mathbb{Z}.$$

□

27th Aug, 2026. Reporting a work in progress for this day and the coming day.

I have been trying to list out all the elements of the automorphism group of a cube explicitly, and here is my attempt at that-link.

26th Aug, 2026. Divisibility and ideals are closely related. In $\mathbb{Z}$, the statement $a \mid b$ means that there exists some $k \in \mathbb{Z}$ such that $b = ak$. Equivalently, $b \in \langle a \rangle$, where $\langle a \rangle = \{ak : k \in \mathbb{Z}\}$ is the principal ideal generated by a. Thus, divisibility by a may be understood as membership in the principal ideal generated by a.

This relationship also explains the connection between divisibility and inclusion of ideals. Suppose that $a \mid b$. Then $b = ak$ for some $k \in \mathbb{Z}$. Every multiple of b is therefore also a multiple of a, and hence $\langle b \rangle \subseteq \langle a \rangle$. Thus divisibility gives an inclusion of principal ideals, although the direction of inclusion is reversed: $a \mid b \Longrightarrow \langle b \rangle \subseteq \langle a \rangle$.

This point of view extends naturally to an arbitrary commutative ring R. For $a \in R$, the principal ideal generated by a is $\langle a \rangle = \{ra : r \in R\}$. Hence, for $a, b \in R$, $a \mid b \Longleftrightarrow b \in \langle a \rangle$. The notion of a principal ideal therefore retains the familiar idea of collecting together all multiples of a single element.

However, ideals are more general than principal ideals. An ideal need not be generated by a single element. Given elements $a_1, \dots, a_n \in R$, the ideal they generate is $(a_1, \dots, a_n) = \{r_1a_1 + \cdots + r_na_n : r_1, \dots, r_n \in R\}$. Thus, while principal ideals arise naturally from divisibility by a single element, general ideals allow several elements to generate a common algebraic object.

In this way, ideals may be viewed as extending the idea of divisibility.

25th Aug, 2026. Let $F_n = \langle x_1, \dots, x_n \rangle$ be the free group on n generators. Since every element of F_n has a unique reduced word, we can describe conjugacy classes in terms of these words.

Two elements $g, h \in F_n$ are conjugate if there exists some $w \in F_n$ such that $h = wgw^{-1}$. Thus, a conjugacy class is $[g] = \{wgw^{-1} : w \in F_n\}$.

Since F_n is a free group, every element has a unique reduced word. A reduced word

$$u = x_{i_1}^{\epsilon_1} \cdots x_{i_k}^{\epsilon_k}$$

is called *cyclically reduced* if its first letter is not the inverse of its last letter. Every element of F_n is conjugate to a cyclically reduced word. Indeed, if the first and last letters of a reduced word cancel when the word is considered cyclically, we can conjugate away these two letters. Repeating this process eventually gives a cyclically reduced word.

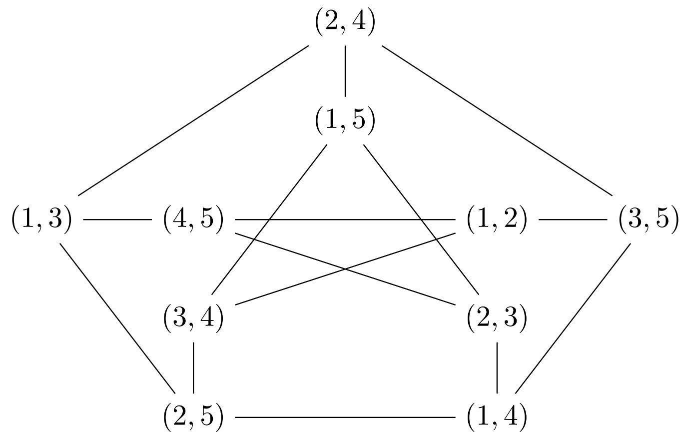
For example, $x_1x_2x_1^{-1}x_2^{-1}$ is already cyclically reduced, whereas $x_1x_2x_1^{-1}$ is not, since its first and last letters are inverses. In this case, $x_1x_2x_1^{-1} = x_1(x_2)x_1^{-1}$, so it is conjugate to the shorter cyclically reduced word x_2.

Now suppose that u and v are cyclically reduced. Then u and v are conjugate iff v is a cyclic permutation of u. For example, $x_1x_2x_1^{-1}x_2^{-1}$ is conjugate to $x_2x_1^{-1}x_2^{-1}x_1$, because the latter is obtained by cyclically permuting the former.

However, that this does not mean that a conjugacy class is finite. The cyclically reduced word only gives us a convenient representative of the conjugacy class. In fact, for $n \geq 2$, every nontrivial conjugacy class in F_n is infinite. For example, in $F_2 = \langle a, b \rangle$, the elements $a, bab^{-1}, b^2ab^{-2}, b^3ab^{-3}, \ldots$ are all conjugate to a, and they are all distinct.

Hence, apart from the identity, the conjugacy classes of F_n can be described by nonempty cyclically reduced words modulo cyclic permutation, even though each such conjugacy class contains infinitely many elements.

24th Aug, 2026. For the sake of *fun*, I tried showing the isomorphisms in Petersen graphs via quiver diagrams. Essentially, all the following graphs are the same up to rearrangement of vertices.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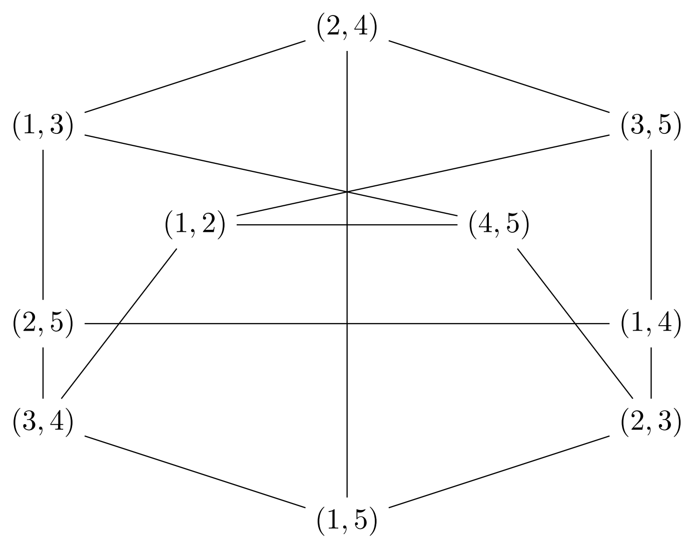
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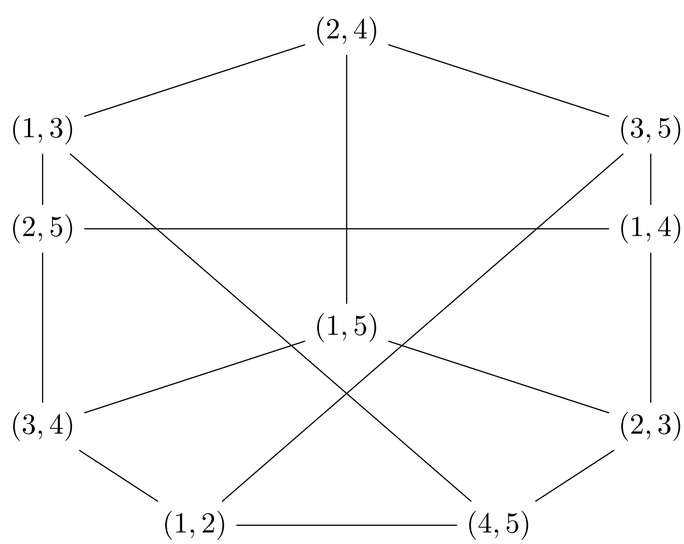
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23rd Aug, 2026. In Fourier theory, the basic objects are characters of a group: homomorphisms from a group G into the circle group $S^1 = \{z \in \mathbb{C} : |z| = 1\}$. Thus a character is a map

$$\chi : G \to S^1 \tag{8}$$

satisfying $\chi(gh) = \chi(g)\chi(h)$. The group operation is translated into multiplication of complex numbers, allowing algebraic information about G to be studied through functions.

This naturally leads to the question of how much information these characters contain about the group. For abelian groups, characters are particularly well behaved: they form the appropriate analogue of the Fourier basis, and functions on the group can be decomposed in terms of them. This is the beginning of character theory, where we study these homomorphisms and the ways in which they reflect the structure of the underlying group. For non-abelian groups, the same philosophy leads us to higher-dimensional representations and, eventually, to the more general notion of a character.

22nd Aug, 2026. Let R be a commutative ring and $I \subset R$ an ideal. An ideal is, in a sense, the basic algebraic object through which we study the structure of a ring, but not all ideals behave equally well with respect to multiplication. The distinction becomes apparent when we consider a product $ab \in I$. For an arbitrary ideal, there is no reason for either a or b to belong to I. However, a prime ideal imposes the strongest possible condition: an ideal $\mathfrak{p} \subset R$ is prime if

$$ab \in \mathfrak{p} \implies a \in \mathfrak{p} \text{ or } b \in \mathfrak{p}. \tag{9}$$

Equivalently, $R/\mathfrak{p}$ is an integral domain. Thus, prime ideals are precisely those ideals for which the quotient has no non-zero zero divisors.

Primary ideals weaken this condition slightly. A proper ideal $Q \subset R$ is called primary if

$$ab \in Q \implies a \in Q \text{ or } b^n \in Q \tag{10}$$

for some $n \geq 1$. Hence a primary ideal allows a product to lie in Q even when neither factor itself lies in Q, provided that one of the factors becomes an element of Q after taking a sufficiently large power.

Every prime ideal is therefore primary, but the converse need not hold. For instance,

$$\langle 4 \rangle \subseteq \mathbb{Z}$$

is not prime, since $2 \cdot 2 \in \langle 4 \rangle$ while $2 \notin \langle 4 \rangle$, but it is primary since whenever $ab \in \langle 4 \rangle$, one of a or b is divisible by 2, and hence a *suitable* power of that element is divisible by 4.

Hence, an ideal Q is primary precisely when every zero divisor in R/Q is nilpotent, whereas a prime ideal gives a quotient which is an integral domain. This provides the useful correspondence

$$Q \text{ primary} \quad \iff \quad \text{every zero divisor of } R/Q \text{ is nilpotent.} \tag{11}$$

21st Aug, 2026. A topological space X is called *irreducible* if it is nonempty and cannot be written as the union of two proper closed subsets, i.e. whenever $X = Z_1 \cup Z_2$ with Z_1, Z_2 closed, then $X = Z_1$ or $X = Z_2$.

Equivalently, every two nonempty open subsets of X have nonempty intersection. Thus, irreducibility is stronger than connectedness, since every irreducible space is connected, but the converse need not hold.

For a topological space X, its dimension is defined as the supremum of the integers n for which there exists a chain of irreducible closed subsets

$$X_0 \subset X_1 \subset \cdots \subset X_n. \tag{12}$$

Thus, $\dim X$ measures the maximum length of a strict chain of irreducible closed subsets. In particular, a point has dimension 0, an irreducible curve has dimension 1, and $\mathbb{A}^n$ has dimension n.

For an affine variety $X = V(I) \subseteq \mathbb{A}^n_k$ with coordinate ring $k[X] = k[x_1, \ldots, x_n]/I$, its dimension is defined by

$$\dim X = \dim k[X]. \tag{13}$$

Equivalently, $\dim X$ is the supremum of the lengths of chains of irreducible closed subsets (as discussed above)

$$X_0 \subset X_1 \subset \cdots \subset X_n. \tag{14}$$

Algebraically, this corresponds to chains of prime ideals in $k[X]$.

For a ring A, the *Krull dimension* $\dim A$ is the supremum of the lengths of chains of prime ideals

$$\mathfrak{p}_0 \subset \mathfrak{p}_1 \subset \cdots \subset \mathfrak{p}_n. \tag{15}$$

The appearance of prime ideals comes from the correspondence between prime ideals of A and irreducible closed subsets of $\operatorname{Spec} A$, where $\mathfrak{p}$ corresponds to $V(\mathfrak{p})$. Thus, the Krull dimension of A agrees with the dimension of its spectrum.

20th Aug, 2026. We describe, in brief, about the *Pontryagin duality*, as an attempt to understand something that comes after.

(1) A *topological group* G is a topological space and a group, such that the group operation is defined as

$$\cdot : G \times G \to G \text{ such that } (x, y) \mapsto xy \tag{16}$$

and the inverse map

$$^{-1}\colon G \to G \text{ such that } x \mapsto x^{-1} \tag{17}$$

are continuous.

We view $G \times G$ as a topological space with the product topology.

(2) Let X be a topological space. X is said to be *locally compact* if every point $x \in X$ has a compact neighbourhood.

(3) A *Hausdorff space* in a topological space where distinct points have disjoint neighbourhoods. We also call this a separated space.
(4) A *locally compact group* is a topological group for which the underlying topology is locally compact and Hausdorff.
(5) For a locally compact abelian group G, the group of continuous homomorphisms $\mathrm{Hom}(G, S^1)$ is locally compact.
(6) Pontryagin Duality says that the following functor induces an equivalence of categories.

$$\mathrm{Cat}(\text{Locally Compact Abelian Groups})^{op} \to \mathrm{Cat}(\text{Locally Compact Abelian Groups}) \tag{18}$$

(7) An *equivalence of categories* consists of the following:
  - two catgeories, say $\mathscr{C}$ and $\mathscr{D}$,
  - two functors, say $F : \mathscr{C} \to \mathscr{D}$ and $G : \mathscr{D} \to \mathscr{C}$, and
  - two natural isomorphisms, say $\eta : FG \to I_{\mathscr{D}}$ and $\varepsilon : I_{\mathscr{C}} \to GF$, where I_{cat} denotes the identity functor omn the respective category.

  Equivalently, a functor is said to establish an equivalence of catgeories if it is full, faithful and essentially surjective.

We will perhaps discuss more about this in the upcoming days, as I try to build some substantial understanding fo the subject.

19th Aug, 2026. For a ring R, R is Noetherian $\iff$ $I_1 \subseteq I_2 \subseteq \cdots$ stabilizes, i.e. R satisfies the ACC on ideals.

For $X = \operatorname{Spec} R$, ideals correspond contravariantly to closed sets:

$$I \subseteq J \quad \Longrightarrow \quad V(J) \subseteq V(I). \tag{19}$$

Thus

$$I_1 \subseteq I_2 \subseteq \cdots \quad \Longrightarrow \quad V(I_1) \supseteq V(I_2) \supseteq \cdots . \tag{20}$$

The order reverses because Spec is contravariant. We then arrive at the following definition.

A topological space X is called *Noetherian* if every descending chain of closed subsets $X_1 \supseteq X_2 \supseteq X_3 \supseteq \cdots$ stabilizes, that is, there exists $n \in \mathbb{N}$ such that $X_n = X_{n+1} = X_{n+2} = \cdots$.

Dually, a topological space X is called *Artinian* if every ascending chain of closed subsets $X_1 \subseteq X_2 \subseteq X_3 \subseteq \cdots$ stabilizes, that is, there exists $n \in \mathbb{N}$ such that $X_n = X_{n+1} = X_{n+2} = \cdots$.

Now, noetherian spaces occur naturally in algebraic geometry, at least, most of the notions are defined on noetherian topological spaces. It is useful because it gives finiteness, such that descending chains of closed sets terminates, every closed subset has finitely many irreducible components, and every open subset is quasi-compact[1].

[1] This is the same as compactness in metric spaces.

Naturally, a question arises to observe the theory of Artinian topological spaces and if it admits any stronger results than noetherian topological spaces.

18th Aug, 2026. An adjunction between categories $\mathscr{C}$ and $\mathscr{D}$ consists of functors $F : \mathscr{C} \to \mathscr{D}$ and $G : \mathscr{D} \to \mathscr{C}$ together with a natural bijection

$$\mathrm{Hom}_{\mathscr{D}}(F(X), Y) \cong \mathrm{Hom}_{\mathscr{C}}(X, G(Y)). \tag{21}$$

We write $F \dashv G$, calling F the left adjoint and G the right adjoint.

The adjunction expresses a universal relationship between the two functors: morphisms out of $F(X)$ correspond naturally to morphisms into $G(Y)$. This correspondence is encoded by the unit and counit of the adjunction and often allows constructions in one category to be translated into another.

A fundamental consequence is that left adjoints preserve colimits, while right adjoints preserve limits.

If $F \dashv G$ and $D : I \to \mathscr{C}$ has a colimit $\operatorname{colim} D$, then

$$F(\operatorname{colim}_{i\in I} D(i)) \cong \operatorname{colim}_{i\in I} F(D(i)) \tag{22}$$

The reason being that, for every $Y \in \mathscr{D}$,

$$\begin{aligned}
\mathrm{Hom}_{\mathscr{D}}\left(F\left(\operatorname{colim}_i \mathscr{D}(i)\right), Y\right) &\cong \mathrm{Hom}_{\mathscr{C}}\left(\operatorname{colim}_i \mathscr{D}(i), G(Y)\right) \\
&\cong \lim_i \mathrm{Hom}_{\mathscr{C}}\left(\mathscr{D}(i), G(Y)\right) \\
&\cong \lim_i \mathrm{Hom}_{\mathscr{D}}\left(F(\mathscr{D}(i)), Y\right) \\
&\cong \mathrm{Hom}_{\mathscr{D}}\left(\operatorname{colim}_i F(\mathscr{D}(i)), Y\right).
\end{aligned} \tag{23}$$

The last equality precisely gives us the universal property of the colimit.

17th Aug, 2026. Some mathematical ideas which I revisited on this day are:

(1) A subgroup generated by a set S is the smallest subgroup containing S. Equivalently, it consists of all finite products of elements of S and their inverses. To prove that S generates a group G, one must show that every element of G can be written as such a product. When proving equality of a generated subgroup with G, both inclusions should be established explicitly. Usually, the inclusion $\langle S\rangle \subseteq G$ is immediate, while $G \subseteq \langle S\rangle$ contains the main argument.

(2) In $\mathbb{Z}/n\mathbb{Z}$, the subgroup generated by $\overline{m}$ is

$$\langle \overline{m}\rangle = \{k\overline{m} : k \in \mathbb{Z}\}.$$

If $\gcd(m, n) = 1$, Bézout's identity gives integers a, b satisfying $am + bn = 1$. Passing to the quotient gives $\overline{a}\,\overline{m} = \overline{1}$. Hence $\overline{1} \in \langle\overline{m}\rangle$. Since every element of $\mathbb{Z}/n\mathbb{Z}$ is an integer multiple of $\overline{1}$, it follows that $\langle\overline{m}\rangle = \mathbb{Z}/n\mathbb{Z}$.

(3) An isometry $f : D \to D$, where $D \subseteq \mathbb{R}$, preserves distances: $|f(x) - f(y)| = |x - y|$ for all $x, y \in D$. Hence, for $x < y$, we must have $f(y) - f(x) = y - x$ for all pairs, or $f(y) - f(x) = -(y - x)$ for all pairs, so f is either increasing or decreasing. Therefore, $f(x) - x$ is constant in the first case, giving $f(x) = x + c$, while $f(x) + x$ is constant in the second, giving $f(x) = -x + c$. Thus every isometry of a suitable subset of $\mathbb{R}$ has one of these two forms.

(4) Stirling's approximation gives the asymptotic behaviour of factorials:

$$n! \sim \sqrt{2\pi n}\left(\frac{n}{e}\right)^n, \qquad n \to \infty. \tag{24}$$

Equivalently

$$\log(n!) = n \log n - n + \frac{1}{2}\log(2\pi n) + o(1). \tag{25}$$

(5) Thomae's function $T : [0, 1] \to \mathbb{R}$ is defined by

$$T(x) = \begin{cases} 1/q, & x = p/q \in \mathbb{Q} \text{ in lowest terms,} \\ 0, & x \notin \mathbb{Q}. \end{cases} \tag{26}$$

The value at a rational depends on the denominator in its reduced form, while T vanishes at every irrational. It is continuous at every irrational point and discontinuous at every rational point. The key idea is that, in any bounded interval, only finitely many rationals have denominator below a fixed bound.

(6) A subset $A \subseteq \mathbb{R}$ is *dense in* $\mathbb{R}$ if $\overline{A} = \mathbb{R}$.

Equivalently, for every $x \in \mathbb{R}$ and every $\varepsilon > 0$, $(x - \varepsilon, x + \varepsilon) \cap A \neq \varnothing$.

Equivalently, every nonempty open interval in $\mathbb{R}$ contains an element of A.

(7) The Baire Category Theorem states that if X is a complete metric space, then X cannot be expressed as a countable union of nowhere dense subsets.

Equivalently, if $X = \bigcup_{n=1}^{\infty} F_n$ with each F_n closed, then at least one F_n has nonempty interior.

Equivalently, if $U_n \subseteq X$ are dense and open for every n, then $\bigcap_{n=1}^{\infty} U_n$ is dense in X.

16th Aug, 2026. Some mathematics which I encountered on thsi day has been listed below.

(1) For a vector space V and a subspace U, the quotient V/U consists of cosets $v + U$. To show that operations on V/U are well-defined, one checks that changing representatives of the cosets does not change the resulting coset. This follows from U being a subspace.

(2) The span of $S = \{v_1, \ldots, v_n\}$ is the set of all linear combinations $\sum_{i=1}^{n} \lambda_i v_i$. The set S spans V if every vector in V can be expressed as such a linear combination.

(3) Vectors $v_1, \ldots, v_n$ are linearly independent if $\sum_{i=1}^{n} \lambda_i v_i = 0$ implies $\lambda_1 = \cdots = \lambda_n = 0$. If there is a non-trivial linear relation, the vectors are linearly dependent.

(4) A basis of a vector space is a set that is both linearly independent and spanning. Thus, both properties must be checked for a proposed basis.
(5) The standard basis of F^n is $\{e_1, \ldots, e_n\}$, where e_i has 1 in the ith coordinate and 0 elsewhere. Every $(a_1, \ldots, a_n) \in F^n$ can be written as $\sum_{i=1}^{n} a_i e_i$.
(6) The basis of a vector space depends on the field of scalars. For example, $\mathbb{C}$ has basis $\{1\}$ over $\mathbb{C}$ but basis $\{1, i\}$ over $\mathbb{R}$.
(7) The polynomial space $\mathbb{R}[x]$ has basis $\{1, x, x^2, \ldots\}$. Every polynomial is a finite linear combination of these monomials, and uniqueness of polynomial coefficients gives linear independence.
(8) A function $f : X \to F$ has finite support if $\mathrm{supp}(f) = \{x \in X : f(x) \neq 0\}$ is finite.
(9) For each $x \in X$, the function δ_x is 1 at x and 0 elsewhere. The set $\{\delta_x : x \in X\}$ is linearly independent since evaluating a finite linear combination at x_j isolates the coefficient of δ_{x_j}.
(10) If a finite spanning set is linearly dependent, one of its vectors can be expressed as a linear combination of the others and removed without changing the span. Repeating this process gives a linearly independent spanning set, hence a basis.
(11) For a group G, right multiplication $R_g(x) = xg$ gives an anti-homomorphism under the usual left-action convention, while $R_g(x) = xg^{-1}$ gives a genuine left group action.

15th Aug, 2026. This day was mostly spent in trying to understand and revist some topics in mathematics (from varied fields). I am summarising some of them below.

(1) For a polynomial p, the equation $p(x) = a$ has only finitely many real solutions unless p is identically a. These finitely many roots divide $\mathbb{R}$ into finitely many intervals on which $p(x) - a$ has a fixed sign.
(2) A space X is connected if it cannot be written as the union of two disjoint nonempty open subsets. An important result is that the continuous image of a connected space is connected.
(3) If a set can be written as $S = \{\gamma(t) : t \in \mathbb{R}\}$ for a continuous map $\gamma : \mathbb{R} \to \mathbb{R}^n$, then S is connected, since $\mathbb{R}$ itself is connected.
(4) When $a^m = b^n,\ \gcd(m, n) = 1$, the coprimality allows a and b to be expressed as powers of a common quantity: $a = t^n,\ b = t^m$.
This is useful for turning an algebraic relation into a one-parameter parametrization.
(5) If X is a finite set with n elements, then every function $X \to \mathbb{R}$ is continuous when X has the discrete topology. Hence $C(X, \mathbb{R}) \cong \mathbb{R}^n$, so its dimension is n.
(6) The divisors of an integer m usually occur in pairs $d, m/d$. The only time a divisor is paired with itself is when $d = m/d$, which gives $m = d^2$. Thus an integer has an odd number of divisors exactly when it is a perfect square.
(7) Double counting means constructing a set of objects and counting that same set in two different ways. Since both counts give the cardinality of the same set, they must be equal.

(8) For a matrix A, if $Av = \lambda v$, then $A^k v = \lambda^k v$. Consequently, for a convergent power series f, $f(A)v = f(\lambda)v$, so the eigenvalues of $f(A)$ are obtained by applying f to the eigenvalues of A.

(9) For a sequence of functions $f_n : X \to \mathbb{R}$, we say that f_n converges pointwise to $f : X \to \mathbb{R}$ if $\forall x \in X,\ \lim_{n\to\infty} f_n(x) = f(x)$.
Equivalently,$\forall x \in X,\ \forall \varepsilon > 0,\ \exists N = N(x, \varepsilon)$ such that for all $n \geq N$, $|f_n(x) - f(x)| < \varepsilon$.

(10) Uniform convergence $f_n \to f$ means $\sup_x |f_n(x) - f(x)| \to 0$. The important distinction from pointwise convergence is that the same bound works for every x. Therefore, uniform convergence can still be used when the point $x = x_n$ depends on n.

(11) For an operator such as $L(f) = 3f + 2f'$, differentiation lowers the degree of a polynomial by one. This allows the coefficients of a desired preimage to be determined recursively from the highest-degree coefficient downwards, which can establish surjectivity.

(12) Two permutations in S_n are conjugate exactly when they have the same cycle type. If $\sigma^3 = e$, every cycle of σ has length 1 or 3, so only finitely many cycle types are possible.

(13) For a matrix A, the infinity norm is $\|A\|_\infty = \max_i \sum_j |a_{ij}|$. Every eigenvalue satisfies $|\lambda| \leq \|A\|_\infty$. Hence if A has nonnegative entries and every row sums to 1, then every eigenvalue has absolute value at most 1.

14th Aug, 2026. Well, I have not been able to find any naturally arising and interesting example of a presheaf which satisfies the uniqueness axiom for sheaves but satisfies the gluability axiom. There is one example where you take the cover of an open set in a topological space X and define the functions to be, say 0 for all $i \in I$, then this would certainly obey the gluability axiom but separability would cause an issue[2].

Also, for a bipartite graph $K_{m,n}$, the automorphism group is isomorphic to the direct product of the symmetric group, $Aut(K_{m,n}) \equiv S_m \times S_n$. But the questions arises: why the direct product?

Moreover, I had a discussion introducing me to the world to Toposes.

13th Aug, 2026. In algebaric geometry, given an exact sequence of presheaves

$$0 \longrightarrow \mathcal{F}_1 \longrightarrow \mathcal{F}_2 \longrightarrow \cdots \longrightarrow \mathcal{F}_n \longrightarrow 0,$$

we show that the following sequence

$$0 \longrightarrow \mathcal{F}_1 U \longrightarrow \mathcal{F}_2 U \longrightarrow \cdots \longrightarrow \mathcal{F}_n U \longrightarrow 0,$$

[2]Please refer to this link for a detailed discussion of this example.

is also exact. Well, this is an iff result and given the definition of a presheaf, which is defined locally, the first exact sequence gives us the second one. We work the other way around by, again, using the definition of presheaves.

12th Aug, 2026. Mathematical ventures on this day included some more reading into scheme-theoretic algebraic notions from Eisenbud, along with some revisitation of linear algebra, particularly diagonal matrices and diagonalizability, and also a result in metric spaces.

A diagonal matrix is a matrix whose entries away from the main diagonal are all zero, i.e.

$$D = \begin{pmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_n \end{pmatrix}.$$

It is interesting that a matrix A is diagonalizable precisely when there is a basis consisting of eigenvectors of A, equivalently when

$$P^{-1}AP = D$$

for some invertible matrix P and diagonal matrix D.

The columns of P are simply the linearly independent eigenvectors of A. If

$$Av_i = \lambda_i v_i, \qquad i = 1, \ldots, n,$$

then we take

$$P = \begin{pmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{pmatrix}, \qquad D = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix}.$$

Since the eigenvectors are linearly independent, P is invertible, and from

$$AP = PD$$

we obtain

$$P^{-1}AP = D.$$

So, essentially, determining P amounts to finding a basis of eigenvectors of A and placing these eigenvectors as the columns of P.

In metric spaces, every finite metric space with n points can be isometrically embedded into $\mathbb{R}^{n-1}$ with the usual Euclidean metric. It is quite interesting to see an abstract finite metric space being realised concretely as a configuration of points in Euclidean space.

Alongside this, $\mathrm{Spec}(R)$ both as the set of prime ideals of R and through the geometric structure given by the Zariski topology is quite interesting.

We can think of

$$\mathrm{Spec}(R) = \{\mathfrak{p} \subseteq R : \mathfrak{p} \text{ is prime}\}$$

as a collection of prime ideals, and also as a geometric object by putting the Zariski topology on it. For an ideal $I \subseteq R$, we consider

$$V(I) = \{\mathfrak{p} \in \mathrm{Spec}(R) : I \subseteq \mathfrak{p}\},$$

and these sets form the closed sets of the Zariski topology. There is also a structure sheaf $\mathcal{O}_{\mathrm{Spec}(R)}$, and together with the underlying topological space, this gives us the affine scheme associated to R.

10th Aug, 2026. Mathematics on this day included some interesting results about groups acting on metric spaces. One such result was that if a group G acts freely by isometries on $\mathbb{R}^n$, equipped with the Euclidean metric, then G must be torsion-free. This is quite interesting since the geometry of the space places an algebraic restriction on the group itself.

In particular, the idea of algebraic sets as zero sets of ideals and the way varieties arise from imposing irreducibility on these sets is interesting.

09th Aug, 2026. Mathematical ventures on this day included some revisits to linear algebra and analysis, for the most part. The theory of matrices is very rich and gives us some very strong results. For instance, the eigenvalues of a matrix whose entries are non-negative and whose row sums are all equal to 1 must all have absolute value at most 1. That is, if $A = a_{ij}$ satisfies

$$a_{ij} \geq 0, \qquad \sum_j a_{ij} = 1,$$

then every eigenvalue λ of A satisfies

$$|\lambda| \leq 1.$$

In analysis, the Bolzano-Weierstrass theorem tells us that every bounded sequence in $\mathbb{R}^n$ has a convergent subsequence, extending that the *Arzelà-Ascoli* theorem tells us that under suitable conditions, a sequence of functions has a *uniformly* convergent subsequence. In particular, *a uniformly bounded and equicontinuous family of continuous functions on a compact space is relatively compact in the uniform topology.*

Also the idea that a finite metric space can be isometrically embedded into some Euclidean space $\mathbb{R}^n$ is particularly interesting. Thus, given a finite metric space X, there exists some n and a distance-preserving map

$$f : X \longrightarrow \mathbb{R}^n.$$

08th Aug, 2026. Mathematics on this day included some readings on the realisation of groups as metric spaces. We start by taking a group G and its generating set S. Now, for any two elements $g_1, g_2 \in G$ we define the *distance* between them to be the word length $g_1^{-1}g_2$ with respect to S. This induces a metric on G, refered to as the *word metric.* Well, to be precise, the word metric is defined as

$$d_S(g_1, g_2) = |g_1^{-1}g_2|_S,$$

where $|g_1^{-1}g_2|_S$ denotes the minimum number of generators from $S \cup S^{-1}$ needed to express $g_1^{-1}g_2$. Moreover, when we think of groups in terms of their Cayley graaphs, this word metric is precisely the *path metric* of the graph.

Another interesting thing about viewing groups as metric spaces is the analogue of isomorphisms which we obtain in the latter, called an *isometry.* An isometry is simply a bijective function, say f defined as:

$$f : (X, d_X) \to (Y, d_Y),$$

such that

$$d_Y(f(x_1), f(x_2)) = d_X(x_1, x_2)$$

for all pairs $x_1, x_2 \in X$.

Another interesting notion is that of the group of symmetries of a metric space X, denoted as $\text{Isom}(X)$. And a group action of a group G on a metric space X would then be the following group homomorphism

$$G \to \text{Isom}(X).$$

So essentially, for each element $g \in G$, the map gives us an isometry $X \to X$.

07th Aug, 2026. Some mathematics discussions[3] on this day included an alternate proof of the *Bolzano–Weierstrass theorem.* The idea was to start with a bounded sequence $(a_n) \subseteq [a, b]$, divide the interval into two parts, and keep the part which contains infinitely many terms of the sequence. One can keep doing this, obtaining nested intervals

$$I_1 \supseteq I_2 \supseteq I_3 \supseteq \cdots$$

whose lengths tend to zero. One can then choose a subsequence by taking a term from each of these intervals, and this subsequence will converge to the unique point determined by the nested intervals. I particularly liked the rather physical analogy used to explain this.

Also, the *Archimedean property* can be realised through the Tortoise and Hare story, and the *Monotone Convergence Theorem* can be visualised by repeatedly partitioning a bounded interval and narrowing down towards a certain point.

[3]Please refer to this webpage and the notes provided there for these ideas.

In another news, an algebraic set can be defined using an ideal rather than a collection of polynomials. Given an ideal $\mathfrak{I} \subseteq k[x_1, \ldots, x_n]$, we can consider

$$V(\mathfrak{I}) = \{a \in \mathbb{A}_k^n : f(a) = 0 \text{ for every } f \in \mathfrak{I}\}$$

.Moreover, the radical of an ideal has some interesting properties that one can play around with in commutative algebra. In particular,

$$V(\mathfrak{I}) = V(\sqrt{\mathfrak{I}}),$$

which is quite interesting since the algebraic set cannot distinguish between an ideal and its radical.

Modules, on the other hand, generally have two ways of being viewed. While I prefer the ring-action approach,

$$R \times M \longrightarrow M,$$

I hope to learn to appreciate the linear algebra approach as well, where modules over rings are considered analogous to vector spaces over a field.

05th Aug, 2026. One of the more surprising ideas I encountered today was that the common notion of the zero set of a collection of polynomials is elevated to a central geometric object in algebraic geometry. Such a set is called an *algebraic set*, while a *variety* is, roughly speaking, an irreducible algebraic set (depending on the convention being used).

I also found it interesting that these objects are studied in affine space ($\mathbb{A}_k^n$). Although affine space has points with coordinates such as $(0, 0)$ or, more generally, $(0, \ldots, 0)$, it is not regarded as having a distinguished origin in the geometric sense. The point $(0, \ldots, 0)$ is then simply another point of the space, and whether it belongs to the zero set of a polynomial depends entirely on the polynomial itself. For instance, if the constant term is nonzero, then $(0, \ldots, 0)$ need not be a zero of that polynomial.

Another pleasing observation was that the Cayley graph of the free group F_2, with respect to its standard free generators, is the infinite 4-regular tree T_4. Starting from any vertex, the tree branches outward in a remarkably symmetric fashion, and after the initial four edges, there are $4 \cdot 3^{n-1}$ new vertices at distance n from the starting point.

Moreover, algebraic sets are interesting objects. Although they can be defined over any field, many of their fundamental properties are studied over algebraically closed fields. There is also the *Zariski topology*, which arises naturally on affine spaces and whose closed sets are precisely the algebraic sets, at least for $n = 1$ and $n = 2$.

04th Aug, 2026. Well, the Cayley graph of a free group with respect to a free basis is an infinite regular tree. This graph has some interesting valencies at each point and it is unlike the finite groups' graphs which contain cycles. In other news, $SL_2(\mathbb{Z})$ has some interesting elements which are also its generators. It would be interesting to explore the geometry of its Cayley graph.

03rd Aug, 2026. Today I learned that automorphisms of a group induce automorphisms of its Cayley graph (provided the generating set is preserved). This is interesting because every automorphism of a group is a permutation of its underlying set, so there is a natural embedding $\mathrm{Aut}(G) \hookrightarrow \mathrm{Sym}(G)$. Thus, the same automorphism acts both as a permutation of the group and as a symmetry of its Cayley graph.